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#2D Virtual Try-On Model Research

Introduction

Virtual Try-On (VTON) systems allow users to digitally visualize how clothes look on a person without physically wearing them. These systems use computer vision and deep learning techniques to transfer garments from a clothing image onto a target person's image while preserving body shape, pose, and garment details.

The evolution of virtual try-on models has progressed from traditional image warping techniques to advanced diffusion-based generative models that produce highly realistic outputs.

1. VITON (Virtual Try-On Network)

Overview

VITON is one of the earliest deep learning-based virtual try-on systems. It generates a new image of a person wearing a selected garment.

Architecture

Input:

- * Person Image
- * Clothing Image

Modules:

1. Clothing-Agnostic Person Representation
2. Coarse Image Generation Network
3. Refinement Network

Output:

- * Person wearing target garment

Advantages

- * First successful virtual try-on framework.
- * Preserves body pose.
- * Simple architecture.

Limitations

- * Low-resolution outputs.
- * Poor preservation of garment details.
- * Unrealistic boundaries around clothes.

VITON: An Image-based Virtual Try-on Network Supplemental Material

Detailed Network Structures

We illustrate the detailed network structure of our encoder-decoder generator in Figure 1, and that of our refinement network in Figure 2.

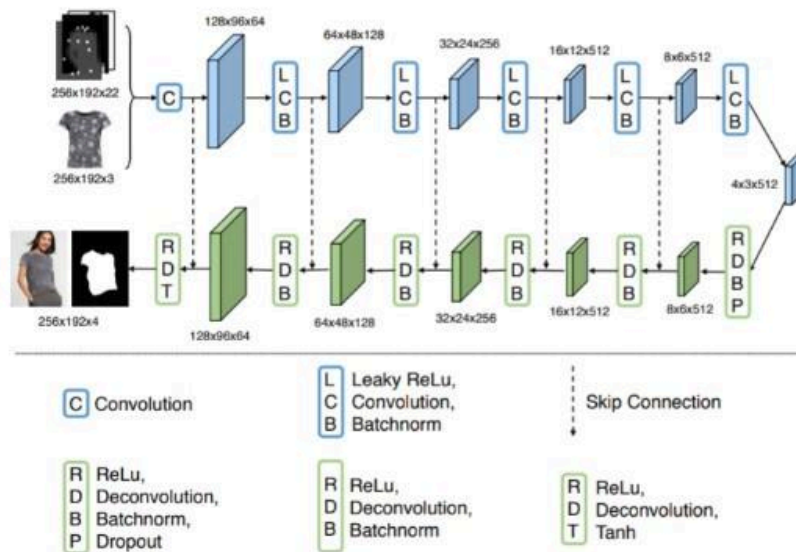


Figure 1: Network structure of our encoder-decoder generator. Blue rectangles indicate the encoding layers and green ones are the decoding layers. *Convolution* denotes 4×4 convolution with a stride of 2. The negative slope of *Leaky ReLu* is 0.2. *Deconvolution* denotes 4×4 convolution with a stride of 1/2. The *dropout* probability is set to 0.5.

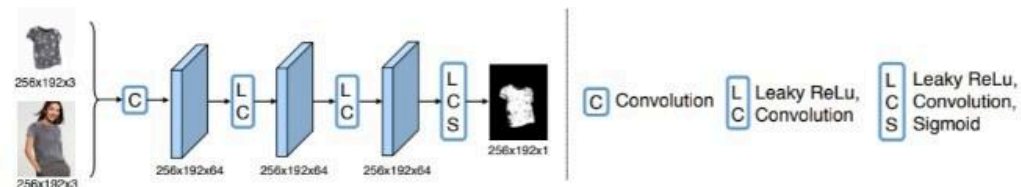


Figure 2: Network structure of our refinement network. *Convolution* denotes 3×3 convolution with a stride of 1. The negative slope of *Leaky ReLu* is 0.2.

2. CP-VTON (Characteristic-Preserving Virtual Try-On)

Overview

CP-VTON improved VITON by introducing a Geometric Matching Module (GMM) for better garment alignment.

Architecture

Input:

- * Person Image
- * Clothing Image

Modules:

1. Geometric Matching Module (GMM)
2. Thin Plate Spline (TPS) Warping
3. Try-On Network

Output:

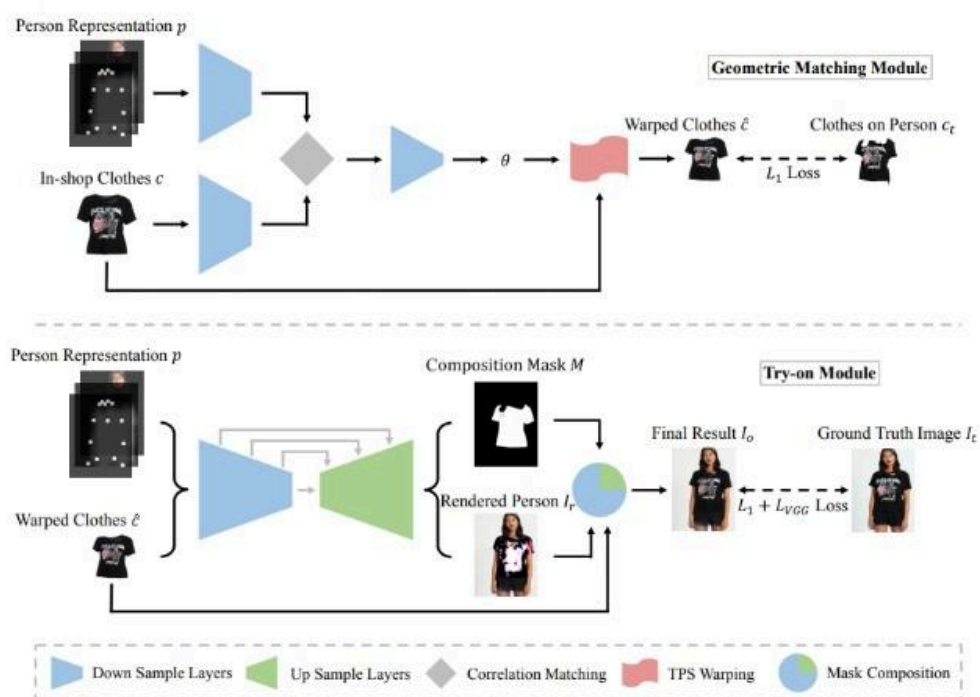
- * Warped garment fitted onto person

Advantages

- * Better garment alignment.
- * Preserves clothing characteristics.
- * Improved realism.

Limitations

- * Struggles with complex poses.
- * Distortion occurs for loose clothing.



3. HR-VITON (High Resolution Virtual Try-On)

Overview

HR-VITON improves image quality and detail preservation compared to previous VITON models.

Architecture

Input:

- * Person Image
- * Clothing Image

Modules:

1. Human Parsing
2. Misalignment-Aware Module
3. High-Resolution Generator

Output:

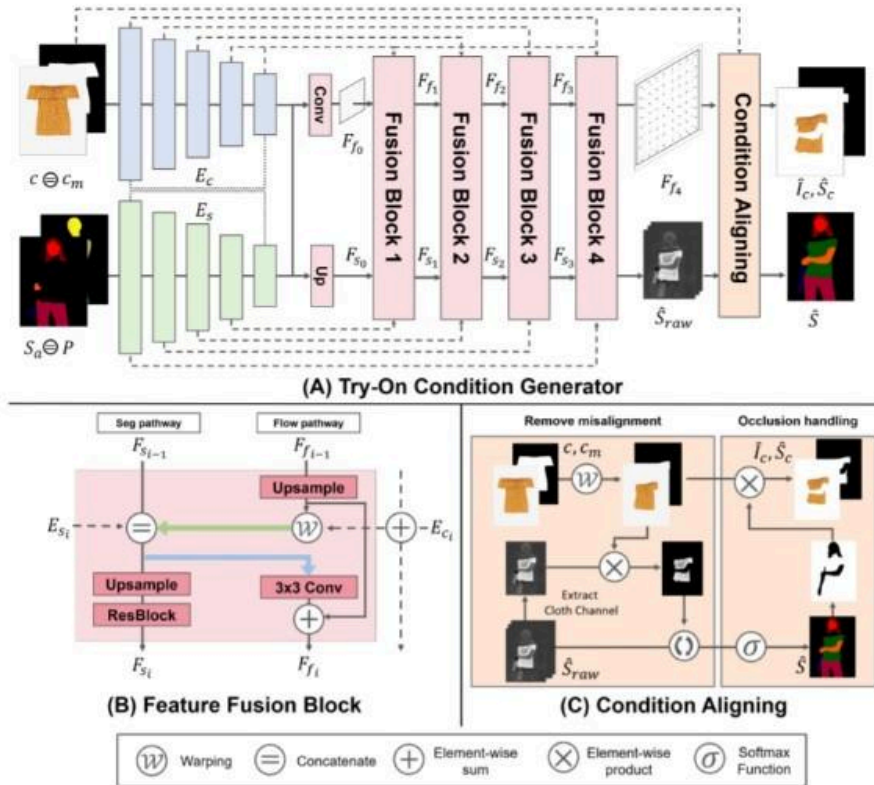
- * High-resolution try-on image

Advantages

- * Better texture preservation.
- * High-quality outputs.
- * Handles complex clothing better.

Limitations

- * Requires more computational resources.
- * Longer training time.



4. StableVITON

Overview

StableVITON combines diffusion models with virtual try-on systems to generate realistic clothing transfer results.

Architecture

Input:

- * Person Image
- * Garment Image

Modules:

1. Stable Diffusion Backbone
2. Garment Encoder
3. Attention Guidance Module
4. Image Decoder

Output:

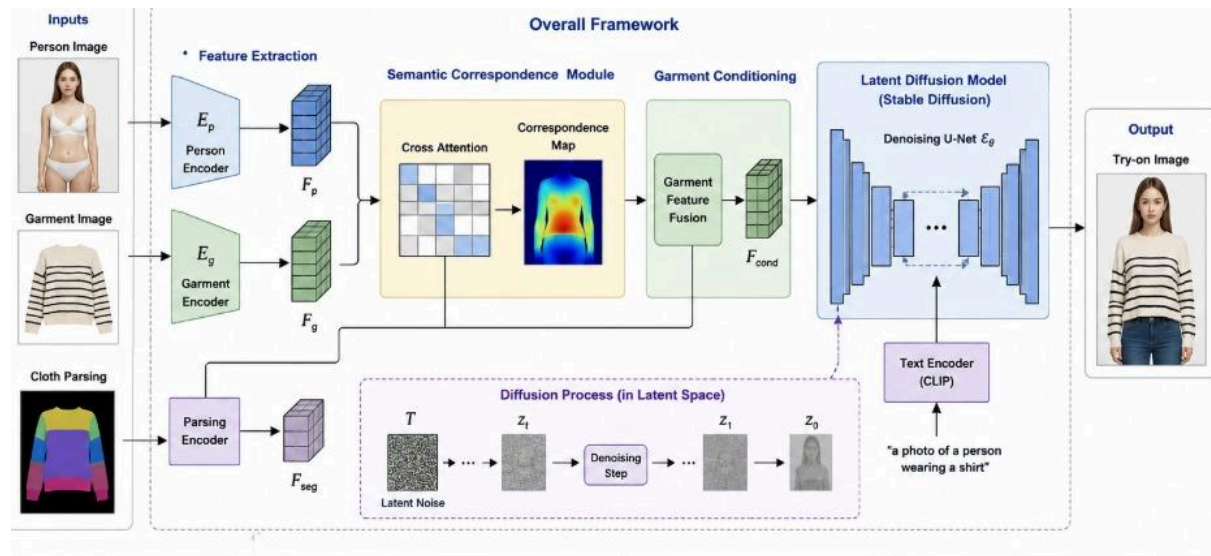
- * Realistic try-on image

Advantages

- * State-of-the-art realism.
- * Better garment texture preservation.
- * Improved clothing fitting.

Limitations

- * High GPU requirements.
- * Slow inference compared to earlier models.



5. IDM-VTON (Improving Diffusion Models for Virtual Try-On)

Overview

IDM-VTON is a diffusion-based virtual try-on model focused on preserving both garment details and human identity.

Architecture

Input:

- * Person Image
- * Garment Image

Modules:

1. Diffusion Model
2. Garment Feature Extractor
3. Identity Preservation Module
4. Attention Mechanism

Output:

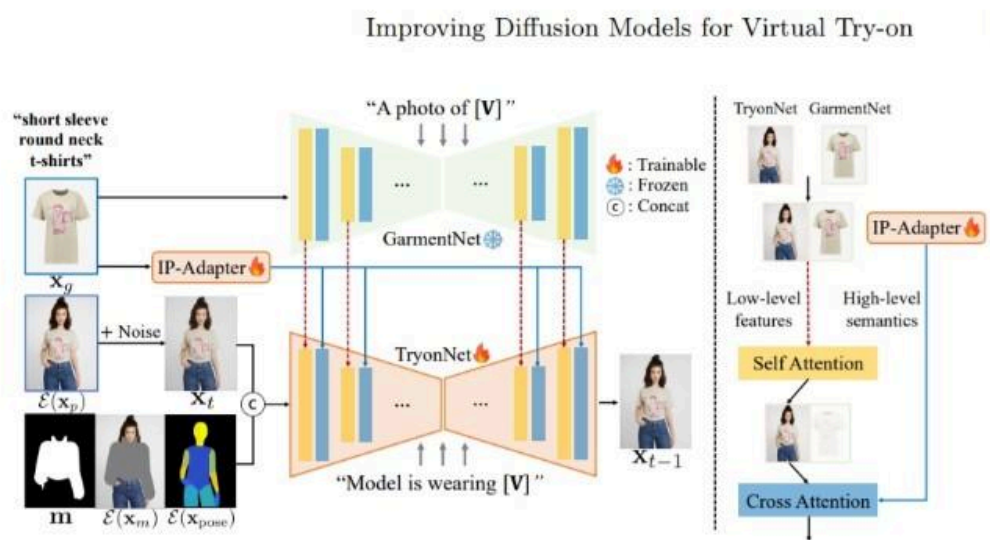
- * High-quality virtual try-on image

Advantages

- * Preserves facial identity.
- * Excellent garment detail retention.
- * Highly realistic results.

Limitations

- * Computationally expensive.
- * Requires powerful hardware.



Model	Year	Technique	Quality	Advantages	Limitations
VITON	2018	Warping + Generation	Medium	Simple architecture	Low realism
CP-VTON	2018	GMM + TPS Warping	Good	Better alignment	Distortion issues
HR-VITON	2021	High-Resolution Gen.	Very Good	Detailed outputs	High computation
StableVITON	2024	Diffusion Model	Excellent	Realistic transfer	Slow inference
IDM-VTON	2024	Diffusion + Identity	Excellent	Best realism	High GPU needs

Advantages and Disadvantages of Different Approaches

Warping-Based Methods

Examples:

- * VITON
- * CP-VTON

Advantages

- * Fast execution.
- * Lower computational cost.
- * Easy to train.

Disadvantages

- * Garment distortion.
- * Poor handling of complex poses.
- * Lower realism.

Diffusion-Based Methods

Examples:

- * StableVITON
- * IDM-VTON

Advantages

- * Highly realistic outputs.
- * Better garment detail preservation.
- * Better fitting performance.

Disadvantages

- * Expensive training.
- * Requires powerful GPUs.
- * Longer processing time.

Findings Summary

1. Virtual try-on technology has evolved significantly from VITON to IDM-VTON.
2. CP-VTON introduced geometric matching, improving garment alignment.
3. HR-VITON improved image quality and clothing detail preservation.
4. StableVITON uses diffusion models to generate highly realistic try-on results.
5. IDM-VTON currently provides some of the best visual quality while preserving both garment details and user identity.
6. Diffusion-based models outperform traditional warping methods in realism but require higher computational resources.
7. Future virtual try-on systems are expected to combine accurate garment fitting, identity preservation, and real-time performance.

References

1. VITON: An Image-based Virtual Try-on Network (CVPR 2018)
2. CP-VTON: Characteristic-Preserving Virtual Try-On Network (ECCV 2018)
3. HR-VITON: High-Resolution Virtual Try-On via Misalignment-Aware Normalization (CVPR 2021)
4. StableVITON (2024)
5. IDM-VTON (2024)